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INTERIM FEDERAL SPECIFICATION

FRANKFURTERS IN BRINE, THERMOSTABILIZED, TRAY PACK

This interim Federal specification was developed by the US Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE

1.1 Scope. This specification covers frankfurters in brine, thermostabilized in tray pack cans for use by the Department of Defense as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

TT-C-495	- Coatings, Exterior, for Tinned Food Cans
PPP-B-636	- Boxes, Shipping, Fiberboard

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification, other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

AMSC N/A FSC 8905

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(Federal Government activities may obtain copies of Federal standardization documents, and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications:

MIL-L-1497	-	Labeling of Metal Cans for Subsistence Items
MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For

Military Standards:

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-900	-	Bacterial Standards for Starches, Flour, Cereals Alimentary Pastes, Dry Milks and Sugars Used in the Preparation of Thermostabilized Foods for the Armed Forces

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Federal Regulations:

U.S. Department of Health and Human Services

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. Department of Agriculture (USDA)

Institutional Meat Purchase Specification for Sausage Products, Series 800

Meat and Poultry Inspection Regulations

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

United States Standards for Condition of Food Containers

(Application for copies should be addressed to the Director, Market Research and Development Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

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Environmental Protection Agency (EPA)

National Interim Primary Drinking Water Regulations

(Application for copies should be addressed to the Office of Drinking Water WH550, Environmental Protection Agency, Washington, DC 20460.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Association of Official Analytical Chemists (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

National Academy of Sciences

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.1, and 6.3)

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Frankfurters. The frankfurters shall conform to item No. 800, formula D product, color A, style C, chilled state of the Institutional Meat Purchase Specification for Sausage Products, except that the frankfurters shall be 5.25 ± 0.25 inches in length and 10 frankfurters shall weigh 1 pound. The frankfurters shall be prepared such as to conform to the requirements specified in 3.3. The frankfurters shall have been held in the temperature range of 30° to 40°F for a period of time not to exceed 14 days from the date of manufacture to the start of tray pack filling (see 3.4) and shall be in excellent condition at time of filling.

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3.2.2 Sugar. Sugar shall be clean, refined, liquid, granulated cane or beet, or a combination of both. The sugar shall comply with MIL-STD-900.

3.2.3 Salt. Salt shall be noniodized, white, refined, sodium chloride with or without anticaking agents.

3.2.4 Pepper, white, ground. Ground white pepper shall be derived from the dried mature berries of Piper nigrum L. from which the outer covering or outer and inner coverings have been removed. The ground white pepper shall have a characteristic penetrating odor, a hot biting pungent flavor, and a light color. The white pepper shall contain not less than 1.0 mL of volatile oil per 100 grams and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 40 sieve.

3.2.5 Red pepper. Ground red pepper shall be derived from ripe fruit of the genus Capsicum frutescens and shall possess the characteristic red to brown-red color with a perceptible sensation of pungency. It shall be uniformly ground to allow for a minimum of 95 percent by weight to pass through a U.S. Standard No. 20 sieve. The Scoville Pungency shall be not less than 40,000 units.

3.2.6 Nutmeg, ground. The ground nutmeg shall be derived from the dried seeds of Myristica fragrans Houtt. The nutmeg shall have a dull brown color. The nutmeg shall contain not less than 7.5 mL of volatile oil per 100 grams and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 20 sieve.

3.2.7 Coriander, ground. Ground coriander shall be derived from Coriandrum sativum L. and shall possess a slight fragrant odor, a pleasant aromatic taste, and a yellowish-brown to tan color. The coriander shall contain not less than 0.3 mL of volatile oil per 100 grams and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 30 sieve.

3.2.8 Sodium nitrite. Sodium nitrite shall comply with the Food Chemicals Codex.

3.2.9 Sodium ascorbate and/or sodium erthythrobate. Sodium ascorbate and sodium erthythrobate both shall comply with the Food Chemicals Codex.

3.2.10 Water. Water used for formulation, ice making, and washing shall conform to the National Interim Primary Drinking Water Regulations.

3.3 Preparation and further processing. Processing shall be on a continuous basis.

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3.3.1 Frankfurter preparation. The frankfurters shall be prepared using the following formulation:

Ingredients	Weight
Beef	100. lbs
Salt	40.0 ozs
Sugar	16.0 ozs
Pepper, white, ground	4.5 ozs
Nutmeg	2.0 ozs
Coriander	2.0 ozs
Red pepper	1/8 ozs
Sodium ascorbate1/	7/8 ozs
Sodium nitrite	1/4 ozs

1/ Sodium ascorbate or sodium erythrobate or a mixture of the two may be used.

The prepared frankfurters prior to canning, shall conform to the following requirements:

a. Fat content of the frankfurters shall be not greater than 22 percent.
b. Salt content of the frankfurters shall be not less than 2.0 percent nor greater than 2.8 percent.
c. Moisture content of the frankfurters shall be not greater than 8 percent as added water.

3.3.2 Brine preparation:

a. Salt and water shall be combined to provide a 2.0 to 2.5 percent brine (salt) solution.
b. Brine shall be made fresh daily and brine remaining at the end of a production day shall be discarded.

3.4 Tray pack filling and sealing. Each tray pack can (see 5.1.1) shall be filled with frankfurters and brine such as to conform to the finished product requirements and to the following requirements:

a. Forty-four frankfurters (weighing not less than 68 ounces) shall be packed in the tray pack can in two layers, each layer consisting of two rows with eleven frankfurters per row. If over crowding of the bottom layer results, one frankfurter from each row of the bottom layer shall be placed end to end between the two top rows.
b. Brine may be filled hot or cold.
c. Frankfurter temperature at time of filling into tray pack can shall not exceed 50°F.
d. Each can shall be hermetically sealed (under a vacuum of not less than 10 inches of mercury) immediately after filling.
e. The filled and sealed tray pack cans shall be in the retort process within 1 hour after sealing.

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3.5 Tray pack thermoprocessing. The filled and sealed tray pack cans shall be thermoprocessed by retorting until a sterilization value (F_o) of not less than 6 has been achieved.

3.6 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insects, insect parts, hair, wood, paper, paint, glass, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, musty, or moldy.

c. There shall be no color foreign to the product.

d. No individual can shall contain less than 102 ounces of product.

e. The average net weight shall be not less than 104 ounces.

f. Frankfurters shall fit into the tray without abnormal distortion.

g. Frankfurters shall not be soft, mushy, or abnormally distended.

h. Frankfurters shall not be split or ruptured more than 1/2 inch in any dimension.

i. Frankfurters shall show no evidence of residual casing material from the skinning operation which exceeds 1/4 inch in any dimension.

3.6.1 Palatability. The finished product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The meat component and the finished product shall originate and be produced, processed, and stored in plants regularly operating under Meat and Poultry Inspection Regulations of the U.S. Department of Agriculture.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

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4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of supervision necessary to assure compliance with the requirements of this specification.

4.3 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

4.4 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, or compliance be evident by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. In addition, prior to use, each ingredient shall be examined organoleptically, as necessary, to determine conformance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.4.2 Frankfurter examination. Examination of frankfurters shall be made to determine compliance with the requirements of 3.2.1. In addition, records of formulation, temperatures, and holding times shall be maintained. When frankfurters are obtained from a contractor other than the prime contractor, the frankfurters shall receive a source inspection to determine compliance with 3.2.1 and shall be so certified by the USDA. Failure to comply with requirements shall be basis for rejection of the lot.

4.4.3 Fat, salt and moisture testing. After completion of the skinning operation, 39 pounds of frankfurters shall be randomly selected from the day's (one work shift) production and distributed as follows:

- Thirteen one pound samples for laboratory analyses
- Thirteen one pound samples for submission to the contractor
- Thirteen one pound samples for retention by an AMS agent as a reserve sample.

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The thirteen one pound samples for laboratory analyses shall individually be tested for fat, salt, and moisture content in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists. The moisture content shall be determined in accordance with the Added Water test method in Section 24.004 of the Meat and Meat Products chapter. The lot shall be rejected if:

a. The fat content of any one pound sample is greater than 22 percent.
b. The salt content in any one pound sample is less than 2.0 percent or greater than 2.8 percent.
c. The moisture content is greater than 8 percent as added water.

Analysis of reserve samples at the request of the contractor shall not be permitted unless the original analysis indicated that the involved lot will be rejected because of non-compliance with the fat, salt, or moisture content requirement. When the reserve samples are analyzed, the analysis for fat, salt and moisture shall be made and will be considered final. Unused reserve samples shall be returned to the contractor for inclusion in subsequent lots.

4.5 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, can interior coating, filling, sealing, and packing requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.6 Tray pack inspection. The inspection lot shall include only tray packs produced in one workshift. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

4.6.1 Net weight inspection. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty tray pack cans and lids used in preparing the product and dividing the total weight by 30) from the weight of each tray pack in the sample. The results shall be reported to the nearest 1 ounce. If the average net weight is less than 104 ounces or if the net weight of any individual can is less than 102 ounces, the lot shall be rejected.

4.6.2 Product inspection. The sample size shall be as indicated by the double sampling plan specified in table I. The tray pack cans shall be selected at random from the lot. The tray pack cans shall be heated for at least 10 minutes in 140° to 150°F water, opened, and inspected for the defects listed in table II.

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TABLE I. Double sampling plan for product inspection. 1/

Lot size (cans)	Sample size (No. of cans)	Cumulative sample	Acceptance number	Rejection number
0-3200	8	-	0	2
	8	16	1	2
3201-35000	13	-	0	3
	13	26	3	4

1/ a. If no defects are found in the first sample, the lot shall be accepted.

b. If the number of defects found in the first sample equals or exceeds the rejection number, the lot shall be rejected.

c. If the number of defects found in the first sample exceeds the acceptance number but is less than the rejection number, the second sample shall be inspected. Defects found in the first and second samples shall be combined and if the number of defects in the cumulative sample equals or exceeds the rejection number, the lot shall be rejected.

TABLE II. Product defects 1/ 2/

Category		Defect
Major	Minor	
101		More than two frankfurters less than 5.0 inches in length.
102		Less than 44 frankfurters per tray pack can.
103		Fit of frankfurters in tray pack can causing abnormal distortion.
104		Presence of soft, mushy, or abnormally distended frankfurters.
105		Presence of frankfurters split or ruptured more than 1/2 inch in any dimension.
	201	Presence of residual casing material from the skinning operation which exceeds 1/4 inch in any dimension.

1/ The presence of foreign material (e.g. glass, dirt, insect parts, hair, wood, metal), foreign odor or flavor (e.g. burnt, scorched, moldy,

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rancid, sour, stale), or foreign color shall be cause for rejection of the lot. (To be determined by a Meat and Poultry Inspection Operations employee.)

2/ Product not equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an AMS agent.)

4.7 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the United States Standards for Condition of Food Containers, except that inspection for labeling shall be in accordance with MIL-L-1497 (see 5.4).

4.8 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement (when applicable) in accordance with the appendix of PPP-B-636. In addition, the following defects shall be classified as follows: Major: (1) National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible: (2) Reinforced with other than nonmetallic strapping or tape. Minor: (1) Other required markings missing, incorrect or illegible; (2) Interior packing with fiberboard liner or pads not as specified; (3) Height of box liner less than inside depth dimension of box by more than 1/8 inch or extends above the top edge of box; (4) Length or width of pads less than box dimensions by more than 1/8 inch; (5) Arrangement or number of cans not as specified.

4.9 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The product shall be packaged in accordance with level A or C as specified (see 6.1).

5.1.1 Level A. A size 1001 by 1206 by 200 tray pack can shall be filled with 104 ounces of the food product, sealed and thermoprocessed as specified in 3.4 and 3.5. The base plate of the can shall be electrolytic chromimum-coated steel and shall be of sufficient temper to protect the product during shipment and storage. The base plate weight of the body and end of each can shall be 90 pounds per base box. The entire inside area of the can shall be coated. Scratches or fractures shall not penetrate through the interior can coating. The interior can coating shall not peel or blister when in contact with the product. The interior can coating shall neither affect nor be affected by the packaged product. A certificate of conformance shall be furnished to certify compliance with the interior can coating requirements. The can shall be coated overall on the outside with a coating conforming to type I of TT-C-495.

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5.1.2 Level C. The product shall be packaged as specified in 5.1.1, except that cans with commercial exterior coating will be acceptable.

5.2 Packing. The product shall be packed in accordance with level A, B or C as specified (see 6.1).

5.2.1 Level A packing. Four cans of product, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, grade V2s of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). The pads shall be not less than 1/8 inch of the full length and width dimensions of the box and shall be fabricated of grade V3c fiberboard. The pads shall be placed between the cans and on the top and bottom of the stacked cans. Each fiberboard box shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament reinforced tape in accordance with the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.1). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Four cans of product, packaged as specified in 5.1, shall be packed as specified in 5.2.1, except the box shall be constructed of grade V3c, V3s, or V4s fiberboard.

5.2.3 Level C packing. Four cans of product, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, class domestic, grade 275 of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box and shall be fabricated of the same material as the box.

5.3 Unit loading. When specified (see 6.1), the product, packed as specified in 5.2.2 and 5.2.3, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified. When unit loads are strapped, the strapping shall be limited to nonmetallic strapping.

5.4 Labeling. Each tray pack can shall be labeled in accordance with MIL-L-1497. In addition, the name of product shall be marked, stamping is permitted, on one 1001 by 200 side of the can. The labeling shall be legible when examined in accordance with 4.7 after submerging the can in boiling water for 30 minutes. Paper labels are not permitted. Cans shall show the following statements:

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TO HEAT

a. To heat in water: Submerge sealed can in boiling water. Bring to a boil, continue boiling for 40 to 45 minutes. Avoid overheating to the point where internal pressure is evident (can shows evidence of bulging). Remove from water and open lid for serving.
CAUTION: Use care when opening as pressure may have been generated within the can.

b. To heat in oven: Either punch several holes in lid or tray can or open can in usual manner leaving the loose lid in place to prevent moisture loss. Place in a 350°F oven for approximately 35 to 40 minutes until contents reach an internal temperature of 165°F.

WARNING: Do not place closed can in oven, as heating will increase internal pressure which may cause the can to burst.

YIELD

Approximately 22 servings of 2 frankfurters each.

5.5 Marking.

5.5.1 Shipping containers. In addition to any special marking required by the contract, shipping containers shall be marked in accordance with MIL-STD-129.

5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078. In addition, the following precautionary marking in capital letters larger than other markings shall be included:

CAUTION: DO NOT STACK PALLETS IN TRANSIT OR MORE THAN TWO HIGH IN STORAGE

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. When a first article is required (see 3.1, 4.3, and 6.3).
c. Provisions for approved preproduction samples (see 3.6.1 and 6.3).
d. Level of packaging and packing required (see 5.1 and 5.2).
e. Type and class of unit load when unit loading is required (see 5.2.1 and 5.3).

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6.2 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should include specific instructions in all procurement documents regarding arrangements for selection, inspection, and approval of the first article.

Preparing activity:

Army - GL

Project No. 8905-AB15

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